METEHAN GÜNEN

Network Administrator

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Metrohan | Linkedin | Website

Summary

Final-year Computer Engineering student specializing in Backend Development, IoT, and real-time systems. Experienced in building scalable backend architectures, integrating embedded devices, and designing end-to-end data pipelines. Proven ability to develop production-ready systems combining cloud, hardware, and AI. Strong focus on performance optimization, system design, and real-world engineering problems.

Professional Experience

Network Administrator, *Pievision* — 04/2026 – Present, İstanbul, Türkiye

- Managing network infrastructure and ensuring system reliability and uptime
- Monitoring system performance and troubleshooting network-level issues
- Supporting deployment and maintenance of backend services and internal tools

Researcher, *i-Lab* — 08/2025 – Present, Istanbul, Türkiye

- Developing AI-powered biomedical and IoT systems with real-time processing constraints
- Designed end-to-end pipelines combining embedded devices, cloud services, and computer vision
- Worked on segmentation and detection systems using YOLO and edge computing approaches
- Collaborated in multidisciplinary teams to prototype and validate research-driven solutions

Intern, *BGTS* — 07/2025 – 08/2025, Istanbul, Türkiye

- Built scalable backend services using Node.js & Express with asynchronous architecture
- Designed REST APIs and improved system performance and maintainability
- Integrated PostgreSQL and implemented structured data handling
- Contributed to system architecture decisions and participated in code reviews

Education

B.Sc. Computer Engineering, *Istinye University* — 08/2022 – 06/2026, Istanbul, Türkiye

Technical Skills

Programming
Python, C, C++, SQL, JavaScript

IoT & Embedded Systems
Raspberry Pi, ESP32, Arduino, Sensor Integration

Tools & Platforms
Docker, Git/GitHub, Linux, Nginx, Prometheus, Grafana

Backend & Databases
Node.js, Express, FastAPI, REST APIs, PostgreSQL, MongoDB

Computer Vision/AI
YOLO, PyTorch, TensorFlow, OpenCV, scikit-learn

Languages

Turkish
Native/Bilingual

English
Proficient

Projects

CERAHIS, *Cerrahi Alet Hasar Tespit ve Raporlama Sistemi*
• Capstone project with Semih Çelenk, advised by Dr. Fevzi Aytaç Durmaz (Istinye University / i-Lab)
• Designed an end-to-end system for automated surgical instrument damage detection using YOLOv11m (mAP@50: ~0.99)
trained on 9,361 images across 13 instrument classes
• Built Raspberry Pi 3B+ + Arducam IMX519 hardware cabinet with a CrowPanel ESP32-S3 7" HMI display; optimized USB Serial
streaming from 0.2 FPS to ~15 FPS real-time preview
• Implemented FastAPI backend with confidence-based decision engine (auto_labeled / needs_review / unsure) and React
dashboard with inventory management and reporting modules
• Integrated QR/Datamatrix barcode reading via image processing for instrument identification
• Conducted FMEA risk analysis; system architecture designed for ISO 13485 and IEC 62304 compliance
Patent application filed (pending) — Türk Patent ve Marka Kurum

MEDHOLO
• TÜBITAK 2209-B with Semih Çelenk industry-supported research project developed at i-Lab in collaboration with Pievision Ar-Ge
• AI-powered hospital triage kiosk using a 3D holographic avatar (Three.js) for contactless, hygienic patient interaction at
emergency entrances
• Implemented a hybrid decision architecture: deterministic reflex module for life-threatening keywords (< 0.01s response) + Local
LLM (Qwen2.5) for complex symptom reasoning
• Built Python/GPU-accelerated VAD → STT → LLM → TTS pipeline with lip-sync avatar rendering; "self-healing" infrastructure
resilient to network outages
• Achieved 100% functional end-to-end prototype (v25.0); future roadmap includes E-Nabız integration, biometric pain-scale
analysis, and multilingual support

TechEventRadar (eventradar.dev), *Turkish Tech Event Aggregator*
• Open-source platform aggregating Turkish tech conferences, meetups, and hackathons